Plume-Traj: Backward Source Attribution and Forward Dispersion

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Introduction

In atmospheric science, particularly in the study of volcanic eruptions, determining the source parameters of an airborne plume is a critical task. The height and timing of emissions of materials such as sulfur dioxide (SO_2) and ash are key inputs to atmospheric transport models used to forecast the plume's trajectory and potential impacts on aviation, air quality, and climate. However, direct measurements of these emission parameters during an eruption or after are often difficult or impossible to obtain.

In modeling, volcanic emissions can be prescribed using time-dependent vertical profiles. A widely used approach to derive vertical emission profiles for volcanic plume prediction is to perform trial-and-error adjustments, comparing satellite-based total column observations of the plume with model simulations that assume different emission heights. Inverse modeling methods are considered a more efficient and versatile approach for obtaining optimal emissions than the trial-and-error method.

Here, we introduce an intermediate method implemented in the developed `Plume-Traj` tool for reconstructing volcanic emissions, using backward-in-time parcel trajectories to infer time-dependent altitudes of volcanic material. The `Plume-Traj` can be used when the satellite measurement captures the enhanced values of SO_2, aerosol Index (AI), or Aerosol Optical Depth (AOD). Volcanic eruption monitoring relies on both source attribution (when and how high the material was injected) and also on forward dispersion (where the material travels after release). Therefore, we also offer a tool for the forward dispersion of volcanic material.

This framework is not limited to volcanic plumes; it is applicable to any tracer or emission field represented in WRF-Chem (e.g., anthropogenic, biomass-burning, dust, or other chemistry/aerosol sources).

1 Methodology

These two instruments require precomputed dimensional meteorological fields computed using the Weather Research and Forecasting (WRF-Chem) model [Skamarock et al., 2005, Grell et al., 2005], which has been recently corrected and updated in terms of simulating the volcanic eruptions [Ukhov et al., 2025]

- **Backward source attribution:** `plume_backtraj.py` starts from a satellite-retrieved column and traces a large number of virtual air parcels backward in time to a receptor vicinity, which is prescribed using a cylinder spanning the whole atmospheric column. By identifying the time and altitude at which these back-trajectories intersect with a cylinder, the script provides an estimate of the time-height profile of emission information.

- **Forward dispersion:** `plume_forwtraj.py` releases a vertical column of parcels at the source and advects them forward to map transport pathways and parcel ages.

Optional physics includes gravitational settling of ash and sulfate aerosols, and SO_2 mass correction with prescribed e-folding lifetime.

1.1 Input Data

Both workflows of the `Plume-Traj` require meteorological data: one or more WRF wrfout NetCDF files (`--wrfout`) containing grid geometry (`XLAT`, `XLONG`, `DX`, `DY`, map factors), winds (`U`, `V`, `W`), geopotential (`PH`, `PHB`), and `Times`. When multiple files are supplied, they are assumed to be ordered in time (or provided via a sortable wildcard).

1.2 Parcel Initialization for backward-in-time run

The backward-in-time simulation begins at a specified time (parameter `--start-time`) by defining the starting locations of the backward-in-time trajectories. The start time must fall within the WRF output time range. This process is called parcel seeding:

- The location of the NetCDF file (`--column`) containing a 2D field representing the total vertical column density (or its equivalent) of some substance (e.g., SO_2 column, AI, or AOD) on the same horizontal grid as the WRF data (Sec. 4). A plume mask is created from the satellite column data (`--column-var`) (optionally scaled by `--column-coef`) by identifying all grid cells where the substance's concentration exceeds a user-defined threshold (`--threshold`). Within this masked area, a specified number (`--n-columns`) of horizontal locations is randomly chosen.
- At each of these locations, a vertical line of parcels (`--n-vert`) is initialized, distributed randomly between a minimum (`--z-min`) and maximum (`--z-max`) altitude. Parcels are placed in the corresponding WRF grid cell. Increasing the total number of parcels (via larger `--n-columns` and/or `--n-vert`) generally reduces sampling noise and yields more stable, converged results, at the cost of longer runtimes.
- Parcels can also be released from columns randomly sampled inside the defined region `--seed-bbox`.

1.3 Parcel Initialization for forward-in-time run

Forward runs require a source location or a seed box:

- `--source-lat`, `--source-lon` for a single column, or `--seed-bbox` with `--n-columns` for random source columns (available in both WRF and MPAS modes when `--emission-matrix` is not used).
- Parcels are initialized between `--z-min` and `--z-max` with `--n-vert` samples. The run spans `--start-time` to `--end-time`.

1.4 Backward/forward Trajectory Calculation

The core of both scripts is the backward/forward advection of air parcels. The position of each parcel is integrated backward/forward in time using a fourth-order Runge-Kutta (RK4) numerical scheme.

$$\frac{d\vec{x}}{dt} = \vec{v}(\vec{x}, t) \tag{1}$$

For the backward mode, the integration is performed as:

$$\vec{x}(t - \Delta t) \approx \vec{x}(t) - \vec{v}(\vec{x}, t)\Delta t \tag{2}$$

The velocity vector $\vec{v}$ for each parcel is determined by interpolating the U, V, and W wind components from the surrounding WRF grid points in both space and time. The parameter `--integration-dt`, in seconds, controls the advection time step. Each parcel carries a weight derived from the column field, enabling both count-based and mass-weighted diagnostics.

1.5 Gravitational settling of aerosols

For aerosol species, the tool can account for gravitational settling by incorporating a size-and altitude-dependent terminal velocity. In WRF-Chem, ash particles are categorized into 10 size bins based on their radii, ranging from 0.01955 to 1000.0 µm, see Table 1. For ash gravitational settling, terminal velocities are calculated for the mean arithmetic radii (R_m) within each bin. Ash density is assumed to be 2500 kg m^{-3}, sulfate density is 1800 kg m^{-3}, and it is assumed that sulfate particles have a dry volume median radius = 0.62 μm. The lookup table (`misc/settling_velocity_data.py`) for different aerosol types (e.g., `vash_1` through `vash_10`, and `sulf`) is being used to apply an additional settling velocity to the parcels during advection. The lookup table values of settling velocity profiles were computed in Ukhov et al. [2025]. For aerosol types (e.g. `sulf`, `vash_1 - vash_10`) a settling velocity profile can be added using `--aer-type`.

Bin	Radius lower bound (µm)	Radius upper bound (µm)	R_m (µm)	Density (kg m^{-3})
vash_1	500	1000	500	2500
vash_2	250	500	375	2500
vash_3	125	250	187.5	2500
vash_4	62.5	125	93.75	2500
vash_5	31.25	62.5	46.88	2500
vash_6	15.625	31.25	23.44	2500
vash_7	7.8125	15.625	11.72	2500
vash_8	3.90625	7.8125	5.86	2500
vash_9	1.95325	3.9065	2.93	2500
vash_10	0.01955	1.95325	0.97	2500

Table 1: Ash particle bin size ranges with corresponding WRF-Chem ash bins.

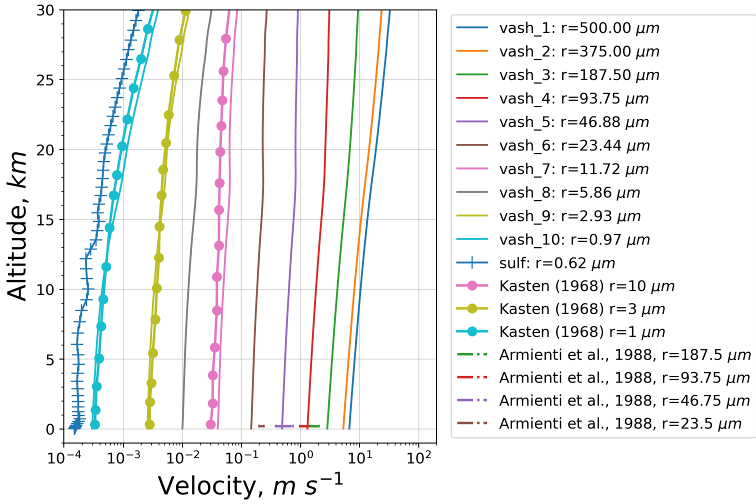

Figure 1: Settling velocity as a function of altitude computed for ash particles with different radii, and for sulfate particles with the dry volume median radii = 0.62 μm. Comparison with theoretical data by Kasten [1968] and Armienti et al. [1988].

Figure 1 illustrates the settling velocities of ash particles as a function of altitude for multiple particle radii. The gravitational settling velocity of sulfate particles having dry volume median radii = 0.62 μm is also shown. We compare our results with theoretical formulations by Kasten [1968] and Armienti et al. [1988]. In Kasten [1968], settling velocities are for spherical particles with radii of 1 μm, 3 μm, and 10 μm; for comparison, the density of these particles was adjusted to align with that of ash. In Armienti et al. [1988], ranges of terminal velocities are provided for feldspar mineral particles (radii of 23.5 μm, 46.75 μm, 93.75 μm, and 187.5 μm), which are regarded as having the closest density to ash particles.

1.6 Source Attribution

To identify the origin of the plume, a cylindrical "receptor" region is defined around the known coordinates of the source (`--receptor-lat`, `--receptor-lon`). This cylinder has a user-defined radius (`--receptor-radius`) and vertical extent (`--receptor-min-h`, `--receptor-max-h`). During the backward integration, the script continuously checks if a parcel has entered this receptor cylinder. If a parcel's trajectory intersects the cylinder, it is considered to have "arrived" at the receptor. `--parcel-radius` (optional, default is 0) sets parcel radius (m) used in receptor-contact checks; a parcel is counted as arrived when the circle touches the receptor cylinder. Parcel arrivals are binned on a time-height grid with `--arrival-bin-minutes`. If both `--emission-start` and `--emission-end` are supplied, fixed time bins spanning the emission window are used, and arrivals outside the window are ignored. The simulation stops for a parcel if it leaves the model domain or reaches the beginning of the emission window (`--emission-start`).
Mass-weighted outputs use the column value multiplied by cell area and divided by `--n-vert`. If `--efolding-days` is set, parcel masses are adjusted backward in time using an exponential correction to compensate for SO_2 oxidation (or radioactive decay, for example).

2 Output

All input parameters and output results can be serialized and stored in a pickle file using `--state-pickle`. The file includes grid data, trajectories, emission matrices, and metadata, allowing plots to be reproduced without re-running the solver. `plot_backtraj.py` and `plot_forwtraj.py` regenerate the same plots from a saved `--state-pickle` file. The aggregation scripts in `misc/aggregate_backtraj.py` and `misc/aggregate_forwtraj.py` combine multiple pickles and output the aggregated plots, for example, for `vash_1` - `vash_10` parcels. All plotting commands accept `--map-extent` to override map bounds when needed and may also include a vertical seed distribution plot (`--seeds-vertical-figure`). `--figure-dpi`: DPI used for all saved figures (default: 200).

For demonstration purposes, the tool was deployed in both modes for the Hayli Gubbi eruption on 23 November 2025. We run backward in time using SO_2 and aerosol index observations from TROPOMI on 24 November at 11:00 UTC. These examples are implemented in the code `https://github.com/saneku/Plume-traj` by default.

2.1 Output from the backward in time run

The script `plume_backtraj.py` generates a series of quantitative and visual outputs, see Fig. 2.

- Emission Time-Height Matrix (`--output-txt`, `--output-figure`): The primary output is a 2D histogram that bins the "arrived" parcels by their arrival time and altitude. This matrix, saved as both a text file and a plot, represents the estimated emission profile from the source, showing the most likely periods and altitudes of significant emissions.
- `--mass-figure`: Optional PNG figure for the mass-weighted time-height emission matrix (same time binning as `--output-figure`).
- Trajectory Plots (`--trajectory-figure`): A map showing the horizontal paths of all parcels that successfully reached the receptor. Trajectories are colored by an attribute, such as their initial altitude, providing a visual link between the observed plume and the source.
- `--trajectory-emission-time-figure`: Optional trajectory map coloured by emission time (hours since the emission reference time; typically `--emission-start` when provided).
- Diagnostic Plots: Several other figures are generated for detailed analysis:
  - `--seeds-figure`: A map of the initial horizontal positions of the seeded parcels, overlaid on the satellite column data.

- `--missed-trajectory-figure`: A map showing the trajectories of parcels that did *not* reach the receptor, which can be useful for diagnosing the influence of the wind field.
- `--trajectory-age`: A plot of trajectories colored by the time taken to travel from the plume to the source.
- `--trajectory-arrival-height-figure`: Trajectories colored by the height at which they arrived at the receptor.
- `--hourly-output-dir`: If set, saves hourly parcel-location snapshots during the backward integration (e.g., `parcel_positions_hour_XXX.png`) into the provided directory.

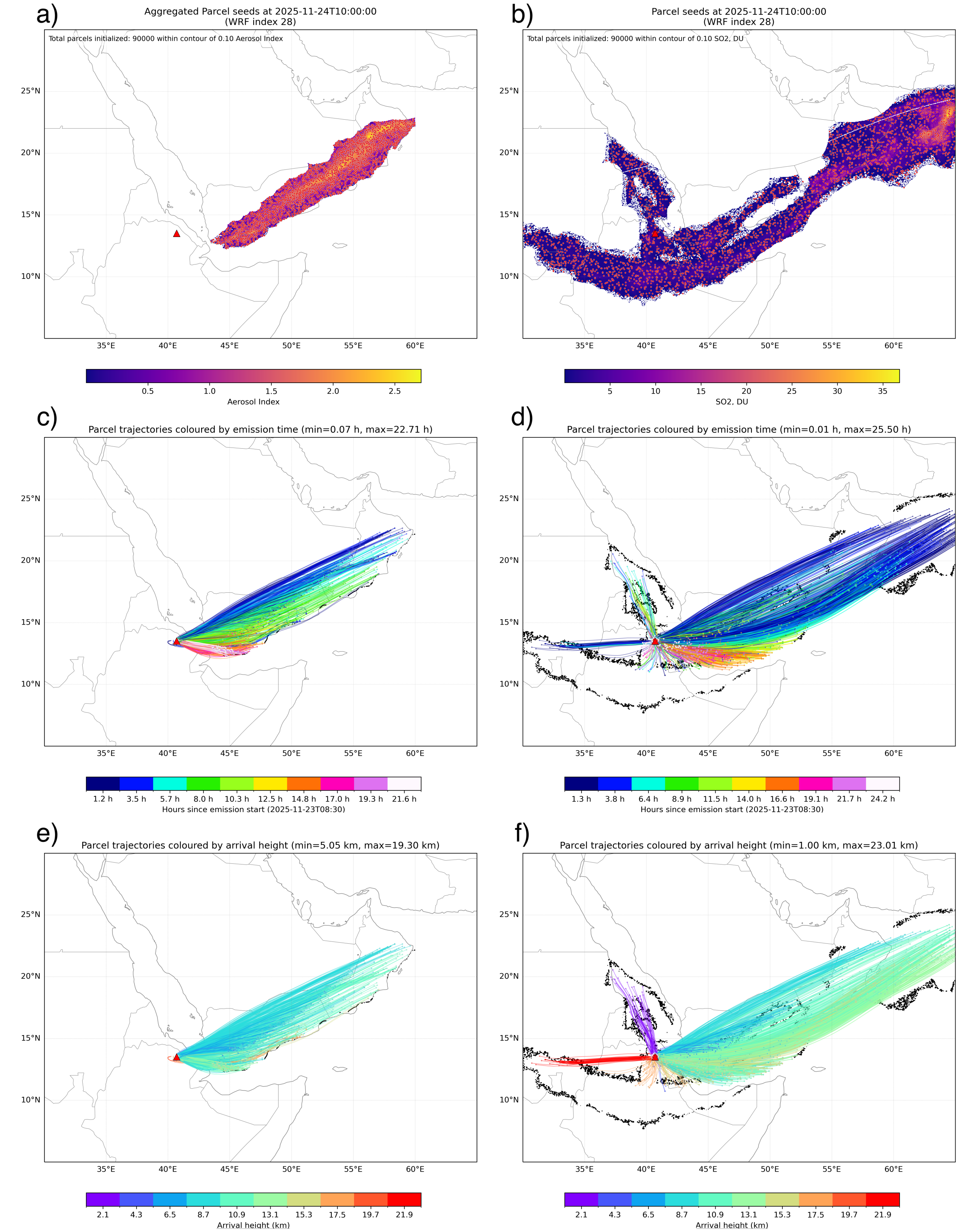

Figure 2: Backward-trajectory reconstruction of the volcanic source using the *Plume-traj* tool (`https://github.com/saneku/Plume-traj`). Ash - left column, SO_2 - right column. Panels a) and b) show the spatial distribution of parcel ”seeds” initialized within the TROPOMI-derived Aerosol Index (AI) a) and TROPOMI SO_2 column loading b) on 24 November 2025, at 11:00 UTC. Panels c) and d) show backward parcel trajectories, shaded according to the number of hours elapsed since the start of the emission, thereby indicating the distribution of arrival times at the site of the Hayli Gubbi volcano. Panels e) and f) show the corresponding arrival heights. Threshold values for ash and SO_2 fields are 0.1 and highlighted by black contour lines. The red triangle marks the Hayli Gubbi volcano.

2.2 Output from the forward in time run

The script `plume_forwtraj.py` is for forward runs and produces height- and age-colored trajectory maps (`--initial-height-figure`, `--age-figure`), as shown in Fig. 3. It can also save hourly parcel-location snapshots (`--hourly-output-dir`) and a deposited-parcel map colored by deposition hour (`--deposition-figure`). For both WRF and MPAS forward runs, map diagnostics use consistent overlays: the source marker (red triangle, from `--source-lat`/`--source-lon`) and the seed area box (`--seed-bbox`, when provided). The same overlay behavior is preserved when re-plotting from `--state-pickle` with `plot_forwtraj.py`.

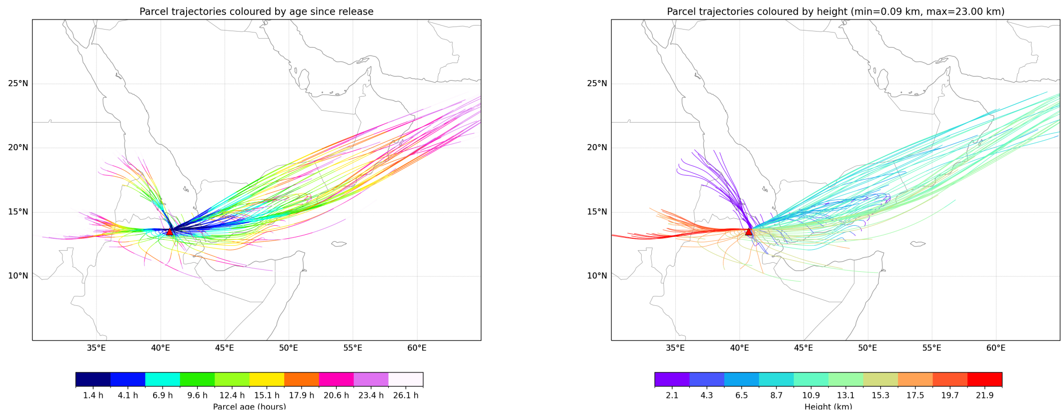

Figure 3: Aggregated forward trajectories for ash (vash_6, ..., vash_10) colored by age (time since the release) (left) and height (right). Computed by forward in time mode using the *Plume-traj* tool (`https://github.com/saneku/Plume-traj`.

3 Coupling to *PrepEmisSources*

The emission matrices obtained in backward in time mode provide a direct bridge to the emission scenarios used by *PrepEmisSources* [Ukhov and Hoteit, 2025]. This links observed plume retrievals to physically consistent emission scenarios that can be injected into WRF-Chem via *PrepEmisSources*. A typical coupling workflow is:

1. Run `Plume-Traj` with `--emission-start` and `--emission-end` aligned to the WRF-Chem simulation window.
2. Use the computed mass-weighted matrix (`--mass-output-txt`) (see Fig. 4) as the time-height distribution of emitted mass;
3. Normalize the matrix to match the desired total emitted SO_2 or ash mass and translate the matrix into the emission scenario format using the *PrepEmisSources*, preserving the time bin edges and vertical structure, see Fig. 5.

4 Pre-processing utilities

- `misc/regrid_wrf.py`: Regrids one or more variables from a source NetCDF grid onto a destination WRF grid using lat/lon coordinates and `scipy.interpolate.griddata`. CLI: `--source-file`, `--dest-file-grid`, `--output-file`, `--variables`, `--time-index`.
- `misc/netcdf_clean_brush.py`: Interactive NetCDF field editor for manual cleaning of 2D products (e.g., SO_2 column/aerosol index) with a brush/eraser tool. Runs on a folder of `.nc` files (positional `folder`) and selects the variable via `--field`; saves edits back into the current NetCDF file.

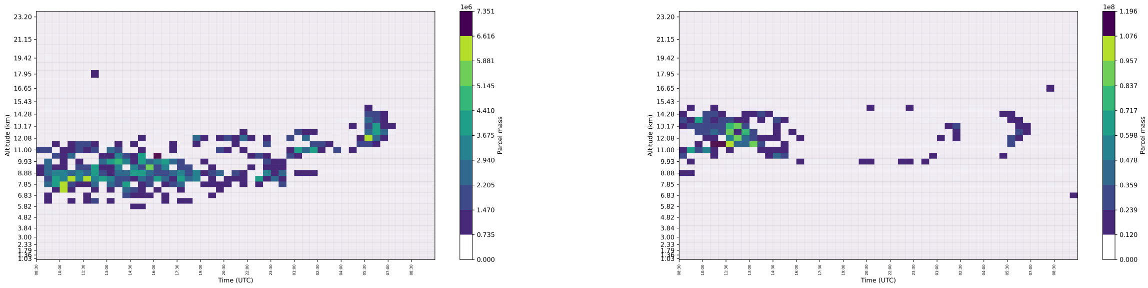

Figure 4: Reconstructed time-height emission matrix based on parcels' age and arrival height, see Fig. 2, respectively. Aggregated ash age parcels (left), SO_2 parcels age (right). Parcels mass in arbitrary units.

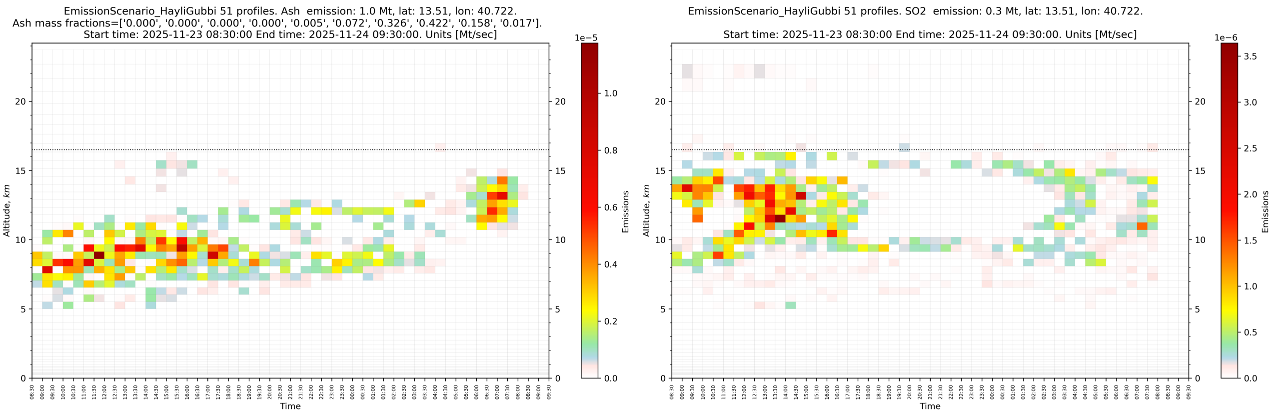

Figure 5: Normalised by erupted mass emission scenarios obtained for ash (left) and SO_2 (right). Plots are constructed using the *PrepEmisSources* utility Ukhov and Hoteit [2025]

References

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